Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/17/2019
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	12/01/2015	**End Date:**	02/29/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G appended with that information per discussions with M. Sansoucie & L. Strutzenberg/MSFC (Ed., 6/21/19) NOTE: Extended to 11/30/2019 per NSSC information (Ed., 12/28/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0256 ; NNX16AB59G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR's NOTE (June 2019): New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G (POP of 12/1/2015-2/28/2019) appended with that information per discussions with M. Sansoucie & L. Strutzenberg/Marshall Space Flight Center (MSFC). Continuation of "Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events" ; grant NNX14AB74G. See that project for previous reports. Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany Subproject: Transformation kinetics - ISS (International Space Station) Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights (PF) on selected alloys. Long Duration Microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS ; Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	• Published five peer-reviewed articles (see Bibliography section) in Acta Materialia, Chemical thermodynamics, and Magnetohydrodynamics. • Graduated two undergraduate BSME Senior Honors Thesis students and two MS students. • Submitted four proposals for 2020 Parabolic Aircraft mission to support samples proposed for Batch 4. • Conducted tests at APS Argonne National Lab on glass-flux samples with inductive heating and stirring in support of Batch 3 and Batch 4 candidate flight samples. • Installed as National President of the American Society for Gravitational and Space Research (ASGSR). • Featured speaker for ISS R&D, ASGSR, International Conference on Advances in Solidification Processes (ICASP-5), and Japan Microgravity Application Society 31st Academic Lecture (JASMAC-31) conferences.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Abstracts for Journals and Proceedings	Krivilyov M, Lomaev S, Fransaer J, Matson DM, Volkmann T. "Numerical simulation of heat and mass transport in the containerless solidification experiment PARSEC onboard of the International Space Station." Solidification – computer simulation, experiments and technology. 8th International Conference, Izhevsk, Russia, April 11-12, 2019. Abstracts of the 8th International Conference Solidification – computer simulation, experiments and technology, Izhevsk, Russia, April 11-12, 2019. p. 75-76. , Apr-2019
Articles in Peer-reviewed Journals	Xiao X, Lee J, Hyers RW, Matson DM. "Numerical representations for flow velocity and shear rate inside electromagnetically levitated droplets in microgravity." npj Microgravity. 2019 Mar 25;5:7. https://doi.org/10.1038/s41526-019-0067-2 ; PubMed PMID: 30937361; PubMed Central PMCID: PMC6433928 , Mar-2019
Articles in Peer-reviewed Journals	Jeon S, Kolbe M, Kaban V, String G, Cleaver A, Kaban I, Shuleshova O, Gao J, Matson DM. "Metastable solidification pathways of undercooled eutectic CoSi-CoSi2 alloys." Acta Materialia. 2019 Sep 1;176(1):43-52. https://doi.org/10.1016/j.actamat.2019.06.048 , Sep-2019

Articles in Peer-reviewed Journals	Jeon S, Sansoucie M, Matson DM. "Hypercooling limit, heat of fusion, and temperature-dependent specific heat of Fe-Cr-Ni melts." Journal of Chemical Thermodynamics. 2019 Nov;138:51-8. https://doi.org/10.1016/j.jct.2019.06.001 , Nov-2019
Articles in Peer-reviewed Journals	Lomaev S, Krivilyov M, Fransaer J, Lee J, Volkmann T, Matson DM. "Simulation of fluid flow in levitated Fe-Co droplets electromagnetically processed onboard of the ISS." Magnetohydrodynamics. 2019;55(1-2):251-9. https://doi.org/10.22364/mhd.55.1-2.30 , Feb-2019
Articles in Peer-reviewed Journals	Matson DM, Xiao X, Jeon S. "Optimization strategies for identifying the controlling mechanism for solid-state transformation in FeCrNi during rapid solidification." IOP Conference Series: Materials Science and Engineering. 2019;529(1):012012. https://doi.org/10.1088/1757-899X/529/1/012012 , Mar-2019
Dissertations and Theses	Priest C. "Radial dependency of droplet oscillatory frequency shift with varying deformation amplitude." Tufts University Mechanical Engineering, BS Honors Thesis, February 2020. , Feb-2020
Dissertations and Theses	Azersky C. "Characterizing the microstructure of Inconel 718 powder using X-ray diffractometry." Tufts University Mechanical Engineering, BS Honors Thesis, May 2019. , May-2019